

# Nils Sturma

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<https://nilssturma.github.io/>

## Academic Employment

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- 09/2025 - today    **Postdoctoral Researcher at EPFL**  
Chair of Biostatistics (Mats Stensrud)  
Also collaborating with the Chair of Mathematical Statistics (Victor Panaretos)
- 10/2024 - 08/2025    **Postdoctoral Researcher at Technical University of Munich**  
Research group Mathematical Statistics (Mathias Drton)

## Education

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- 04/2021 - 09/2024    **Ph.D. in Mathematical Statistics**  
Technical University of Munich (TUM), Germany  
Advisor: Mathias Drton  
Thesis: Identifiability and Statistical Inference in Latent Variable Modeling  
Grade: summa cum laude; date of defense: September 24, 2024.
- 09/2022 - 12/2022    **Research stay at MIT/ Broad Institute in Cambridge, USA**  
Working with Caroline Uhler  
Project: Unpaired Multi-Domain Causal Representation Learning
- 10/2018 - 01/2021    **Master in Mathematical Finance and Actuarial Science**  
Technical University of Munich (TUM), Germany  
Thesis: Testing Algebraic Constraints on Statistical Parameters  
Final grade: 1.2, with distinction (scale: 1 best, 6 worst)
- 02/2020 - 07/2020    **Semester abroad at University of Melbourne, Australia**
- 10/2014 - 01/2018    **Bachelor in Mathematics**  
Albert-Ludwigs-University of Freiburg, Germany  
Thesis: Formal Group Laws  
Minor: Management  
Final grade: 1.1 (scale: 1 best, 6 worst)
- 09/2016 - 02/2017    **Semester abroad at Universidad de Sevilla, Spain**

## Research Interests

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Causality, Graphical Models, High-dimensional Statistics, Algebraic Statistics.

## Publications and Preprints

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- 10 Benjamin Hollering, Pratik Misra, **Nils Sturma**\*. Efficient Symbolic Computations for Identifying Causal Effects. *Submitted*, <https://arxiv.org/abs/2604.20516>.
- 9 Shiva Madadkhani, **Nils Sturma**, Mathias Drton, Svetlana Ikonnikova. Mapping the Causal Structure of Price Formation in Texas's Transitioning Electricity Market. *Preprint*, <https://arxiv.org/abs/2604.14257>.
- 8 **Nils Sturma**, Mathias Drton. Trek-Based Parameter Identification for Linear Causal Models With Arbitrarily Structured Latent Variables. *Submitted*, <https://arxiv.org/abs/2507.18170>.
- 7 Dennis Leung, **Nils Sturma**. Singularity-Agnostic Incomplete U-statistics for Testing Polynomial Constraints in Gaussian Covariance Matrices. *Submitted*, <https://arxiv.org/abs/2401.02112>.
- 6 **Nils Sturma**, Miriam Kranzlmüller, Irem Portakal, Mathias Drton. Matching Criterion for Identifiability in Sparse Factor Analysis. *Psychometrika*, 2026, Vol. 91, No. 2, 536-555.
- 5 Mathias Drton, Alexandros Grosdos, Irem Portakal, **Nils Sturma**\*. Algebraic Sparse Factor Analysis. *SIAM Journal on Applied Algebra and Geometry*, 2025, Vol. 9, No. 2, 279-309.
- 4 Yulia Alexandr, Jane Coons, **Nils Sturma**\*. Mixtures of Discrete Decomposable Graphical Models. *Algebraic Statistics*, 2024, Vol. 15, No. 2, 269-293.
- 3 **Nils Sturma**, Mathias Drton, Dennis Leung. Testing Many Constraints in Possibly Irregular Models Using Incomplete U-Statistics. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 2024, Vol. 86, No. 4, 987-1012.
- 2 **Nils Sturma**, Chandler Squires, Mathias Drton, Caroline Uhler. Unpaired Multi-Domain Causal Representation Learning. *Advances in Neural Information Processing Systems 36, NeurIPS 2023, Spotlight*.
- 1 Rina Foygel Barber, Mathias Drton, **Nils Sturma**\*, Luca Weihs. Half-Trek Criterion for Identifiability of Latent Variable Models. *Annals of Statistics*, 2022, Vol. 50, No. 6, 3174-3196.

\* Shared first authorship/ alphabetical order of the authors.

## Awards/ Grants

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|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Award for Good Teaching   | Third place for best exercise session (High-dimensional Statistics, 2024). Awarded by the Mathematics Student Council of the TUM. |
| SIAM Student Travel Award | Competitive travel award for the 2023 SIAM Conference on Applied Algebraic Geometry.                                              |

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|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MDSI/Linde PhD Fellowship | Competitive grant awarded to PhD students at TUM working on topics related to data science. The grant financed 50% of the 1.0 TV-L E13 position I held during my PhD studies. |
| Alumni-prize 2018         | Every year the prize is awarded by Alumni Freiburg e.V. to the two most outstanding theses (Bachelor or Master) at the Faculty of Mathematics at the University of Freiburg.  |

## Teaching Experience

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|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lecturer            | Randomization and Causation at EPFL, course for students at the end of their Bachelor/ beginning of their Master, spring term 2026 (approx. 60 students).                                                                                                                                                                                                                                           |
| Substitute Lecturer | In total four lectures of the master course Graphical Models at TUM (summer terms 2023 and 2025). One lecture of the master course High-dimensional Statistics at TUM (summer term 2024).                                                                                                                                                                                                           |
| Teaching Assistant  | Graphical Models at TUM, summer term 2025.<br>High-dimensional Statistics at TUM, summer term 2024.<br>Linear Algebra 2 at University of Freiburg, summer terms 2016 and 2017.                                                                                                                                                                                                                      |
| Master Seminars     | Advances in Statistical Inference at TUM, winter term 2024/2025.                                                                                                                                                                                                                                                                                                                                    |
| Mentorship          | Orso Renucci, Bachelor project, 2026.<br>Jannis Friebe, Master thesis, 2025.<br>Daniels Birmans, Bachelor thesis, 2025.<br>Mingyi Guo, Master thesis, 2025.<br>Carolina Kornitzer, Bachelor thesis, 2025.<br>Miriam Kranzlmüller, Interdisciplinary Project, 2024.<br>Javier Yraola Meins, Bachelor thesis, 2023.<br>Moritz Ebert, Master thesis, 2023.<br>Julian Rittmaier, Bachelor thesis, 2022. |
| Upskilling          | “ProLehre” Onboarding Course offered by TUM: Designed to prepare instructors for their teaching task. Elective Module: Supervising Students’ Theses.                                                                                                                                                                                                                                                |

## Professional Activities

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- Reviewer for Annals of Statistics, Biometrika, Electronic Journal of Statistics, Journal of Machine Learning Research, Bernoulli, Algebraic Statistics, La Matematica, NeurIPS conference, UAI conference, CLear conference.
- Part of the organizing committee for the 2025 *CLear conference*.

- Co-organizer of the Workshop *Causal Inference for Time Series Data* at UAI 2024.
- Co-organizer of the *European Workshop on Algebraic Statistics and Graphical Models* 2024.
- Co-organizer of the mini-symposium *Algebraic Methods in Graphical Models* at the SIAM Conference on Applied Algebraic Geometry 2023.

## Software

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- **SEMID**: An R-package that provides routines to check identifiability of causal effects in linear models. Developer and maintainer on [CRAN](#).
- **TestGGM**: An R package for testing the goodness-of-fit of Gaussian graphical models. Lead developer, available on [GitHub](#).

## Talks & Presentations

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| 04/2026 | European Causal Inference Meeting, Oxford, UK.<br>Poster presentation on <i>Trek-Based Parameter Identification for Linear Causal Models With Arbitrarily Structured Latent Variables</i> . |
| 02/2026 | Statistics Seminar at Binghamton University, USA.<br>Invited talk on <i>Matching Criterion for Identifiability in Sparse Factor Analysis</i> .                                              |
| 12/2025 | IMS International Conference on Statistics and Data Science, Seville, Spain.<br>Contributed talk on <i>Matching Criterion for Identifiability in Sparse Factor Analysis</i> .               |
| 02/2025 | Statistics Seminar at University College Dublin, Ireland.<br>Invited talk on <i>Identifiability in Sparse Factor Analysis</i> .                                                             |
| 08/2024 | Bernoulli-IMS World Congress in Probability and Statistics, Bochum, Germany.<br>Invited talk on <i>Identifiability in Sparse Factor Analysis</i> .                                          |
| 03/2024 | Statistics Seminar at University of Melbourne, Australia.<br>Invited talk on <i>Identifiability in Sparse Factor Analysis</i> .                                                             |
| 12/2023 | NeurIPS, New Orleans, USA.<br>Poster presentation on <i>Unpaired Multi-Domain Causal Representation Learning</i> .                                                                          |
| 07/2023 | SIAM Conference on Applied Algebraic Geometry, Eindhoven, Netherlands.<br>Contributed talk on <i>Introduction to Algebraic Methods in Graphical Models</i> .                                |
| 04/2023 | Workshop on Causal Representation Learning, Tübingen, Germany.<br>Contributed talk on <i>Unpaired Multi-Domain Causal Representation Learning</i> .                                         |
| 03/2023 | YES Causal Inference Workshop, Eindhoven, Netherlands.<br>Poster presentation on <i>Parameter Identifiability in Latent Variable Models</i> .                                               |

- 03/2023      German Probability and Statistics Days, Essen, Germany.  
Contributed talk on *Testing Many and Possibly Singular Polynomial Constraints*.
- 08/2022      17. Doktorand:innentreffen der Stochastik, Klagenfurt, Austria.  
Contributed talk on *Half-Trek Criterion for Identifiability of Latent Variable Models*.
- 06/2022      IMS Annual Meeting in Probability and Statistics, London, UK.  
Contributed talk on *Half-Trek Criterion for Identifiability of Latent Variable Models*.

## Industry Experience

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- 09/2018 - 02/2021    **Part-time internship in the Data Analytics department at Zeppelin**  
Development of machine learning applications for the construction industry.  
Focus on predictive maintenance, demand forecasting and process optimization.
- 04/2019 - 08/2019    **BMW & TUM Data Innovation Lab (project)**  
Deep Learning approach to predict lane changes using vehicle sensor data.
- 01/2017 - 03/2021    **Freelancer at Ernst-Klett-Verlag**  
Proofreading of mathematical school books.
- 02/2018 - 07/2018    **Internship at BMW**  
Controlling: inner-year targets of entire BMW Group.